

If you use an online resource to solve any problem, please appropriately cite that source; this includes any AI agent/tool such as ChatGPT, Claude, Gemini, etc.

- a. All hw should be uploaded to canvas as a *pdf*. Make sure that if you scan your handwritten notes that they are legible and appropriately oriented.
 - b. Homework is always due on Wednesday after its released at 11:59PM; however, submitting by Monday at 11:59PM will get you extra credit.
 - c. All homework will be self-graded, with the exception of one problem (selected uniformly at random) that will be graded.
 - d. Self-grading is required to get credit and it is a way to get full credit for your homework (up to any mistakes on the random grading of one problem).
 - e. Bonus problems are not required for credit, but they are worth extra credit.
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Problems

Problem 1. (Logistic Function Properties.) The logistic (sigmoid) function is

$$\sigma(z) = \frac{1}{1 + e^{-z}}.$$

- a. Show that $0 < \sigma(z) < 1$ for all real z .
- b. Show that $\sigma(-z) = 1 - \sigma(z)$.
- c. Compute $\sigma'(z)$ and show that

$$\sigma'(z) = \sigma(z)(1 - \sigma(z)).$$

Problem 2. (Binary Logistic Regression.) Consider binary logistic regression with data $\{(x_i, y_i)\}_{i=1}^n$, where $x_i \in \mathbb{R}^d$ and $y_i \in \{0, 1\}$. The logistic model is

$$p(y = 1 \mid x; \theta) = \sigma(\theta^\top x), \quad \text{where } \sigma(z) = \frac{1}{1 + e^{-z}}.$$

The (unregularized) negative log-likelihood is

$$\ell(\theta) = - \sum_{i=1}^n \left[y_i \log(\sigma(\theta^\top x_i)) + (1 - y_i) \log(1 - \sigma(\theta^\top x_i)) \right].$$

- a. Derive an expression for the gradient $\nabla_{\theta} \ell(\theta)$.
- b. Derive an expression for the Hessian $\nabla^2 \ell(\theta)$.
- c. Show that $\ell(\theta)$ is convex.

Hint: you may want to consider one point i.e. $n = 1$, and then generalize.

Problem 3. (Regularized Logistic Regression.) Consider ℓ_2 -regularized logistic regression problem:

$$\min_{\theta \in \mathbb{R}^d} J(\theta) = \ell(\theta) + \frac{\lambda}{2} \|\theta\|_2^2,$$

where $\ell(\theta)$ is the negative log-likelihood from **Problem 2** and $\lambda \geq 0$.

- Derive the gradient $\nabla_{\theta} J(\theta)$.
- Explain qualitatively what happens to the learned parameter θ as $\lambda \rightarrow 0$ and as $\lambda \rightarrow \infty$.

Problem 4. (Decision Boundary in Logistic Regression.) Consider logistic regression in $\mathbb{R}^2$ with parameters $\theta = (\theta_0, \theta_1, \theta_2)$. The model is

$$p(y = 1 \mid x) = \sigma(\theta_0 + \theta_1 x_1 + \theta_2 x_2),$$

where $x = (x_1, x_2)$.

- Show that the decision boundary defined by $p(y = 1 \mid x) = 0.5$ is a straight line in the (x_1, x_2) -plane.
- Write the equation of this line explicitly in terms of $\theta_0, \theta_1, \theta_2$.
- Suppose we augment the feature vector with a quadratic feature x_1^2 , i.e., consider the model

$$p(y = 1 \mid x) = \sigma(\theta_0 + \theta_1 x_1 + \theta_2 x_2 + \theta_3 x_1^2).$$

Describe qualitatively the shape of the decision boundary now.

Problem 5. (A Mathematical View of Bias and Variance.) Let the true model be linear:

$$y = w^{\star \top} x,$$

where $x \in \mathbb{R}^d$ is fixed and known. Suppose we collect n training samples

$$y_i = w^{\star \top} x_i + \varepsilon_i,$$

where the noise terms satisfy

$$\mathbb{E}[\varepsilon_i] = 0, \quad \mathbb{E}[\varepsilon_i \varepsilon_j] = \sigma^2 \delta_{ij} = \begin{cases} 0, & i \neq j \\ \sigma^2, & i = j \end{cases}.$$

Note that $\mathbb{E}[\cdot]$ is linear, i.e., $\mathbb{E}[aX + bY] = a\mathbb{E}[X] + b\mathbb{E}[Y]$. Let $X \in \mathbb{R}^{n \times d}$ have rows $x_i^{\top}$ and $y = (y_1, \dots, y_n)^{\top}$. Consider two estimators:

$$\hat{w}_{\text{LS}} = (X^{\top} X)^{-1} X^{\top} y, \quad \hat{w}_{\lambda} = (X^{\top} X + \lambda I)^{-1} X^{\top} y,$$

where $\lambda \geq 0$.

- Compute the **bias** of $\hat{w}_{\lambda}$, defined as

$$\text{Bias}(\hat{w}_{\lambda}) = \mathbb{E}[\hat{w}_{\lambda}] - w^{\star}.$$

- b. Compute the **covariance** (or variance matrix)

$$\text{Var}(\hat{w}_\lambda) = \mathbb{E}\left[(\hat{w}_\lambda - \mathbb{E}[\hat{w}_\lambda])(\hat{w}_\lambda - \mathbb{E}[\hat{w}_\lambda])^\top\right].$$

- c. Show that the **mean squared prediction error** at a test point x decomposes as

$$\mathbb{E}[(x^\top \hat{w}_\lambda - x^\top w^*)^2] = \underbrace{x^\top \text{Var}(\hat{w}_\lambda) x}_{\text{variance term}} + \underbrace{(x^\top \text{Bias}(\hat{w}_\lambda))^2}_{\text{bias}^2}.$$

- d. Using the above formulas, explain mathematically why increasing λ

- increases bias, and
- decreases variance.

Problem 6. (Interpreting Logistic Regression Coefficients in 1D.) Consider 1D logistic regression with feature $x \in \mathbb{R}$:

$$p(y = 1 \mid x) = \sigma(\theta_0 + \theta_1 x),$$

where $\sigma(z) = \frac{1}{1+e^{-z}}$. That is the probability of $y = 1$ given x , is given by the sigmoid function. The log-odds is given by the logit function, and in the context of logistic regression, the *odds* that $y = 1$ given the feature x is

$$\text{odds}(x) = \frac{p(y = 1 \mid x)}{1 - p(y = 1 \mid x)}.$$

- a. Show that the *log-odds* (logit) is affine in x : that is, show that

$$\log\left(\frac{p(y = 1 \mid x)}{1 - p(y = 1 \mid x)}\right) = \theta_0 + \theta_1 x.$$

- b. Suppose $\theta_1 > 0$. Show that increasing x by Δ multiplies the odds by a factor of $e^{\theta_1 \Delta}$.
- c. If $\theta_1 = 0.7$, what is the factor by which the odds change when x increases by 2 units?